Abstract #00071 ISMRM ABSTRACT

Data-driven relationship between deep brain stimulation lead trajectories and clinical outcomes in Parkinson's disease

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Eric Cito [1], **Samantha Chan**[1,2], **Devin Schoen**[1], **Skyler Deutsch**[1], **Juhi Mehta**[1], **Anoushka Shah**[1], **Sarah S Wang**[3], **Jill L Ostrem**[3], **Deborah Cahn**[4], **Caroline A Racine**[4], **Doris D Wang**[4], **Melanie A Morrison**[1]

[1]Department of Radiology and Biomedical Imaging, University Of California, San Francisco, San Francisco, United States of America
[2]Epidemiology and Biostatistics, University Of California, San Francisco, San Francisco, United States of America
[3]Department of Neurology, University Of California, San Francisco, San Francisco, United States of America
[4]Department of Neurological Surgery, University Of California, San Francisco, San Francisco, United States of America

Presenting Author: *Eric Cito (eric.cito@ucsf.edu)*

Impact

The effect of deep brain stimulation (DBS) leads penetrating non-target brain structures en route to the surgical target may contribute to observed variability in patient outcomes. Understanding whether lead trajectory influences clinical response could improve surgical planning and prognostic assessment.

Synopsis

Motivation: Understanding whether DBS lead trajectory influences clinical response could improve surgical planning and prognostic assessment.
Goals: Develop and validate a data-driven approach to identify brain regions intersected by leads and associate trajectories with change in Parkinson's medication requirements and cognitive outcomes after DBS.
Approach: Leads from 100 individuals were segmented on postoperative imaging and mapped to an atlas. Bootstrapped LASSO selected stable features, and ridge regression estimated outcome effects alongside clinical covariates.
Results: Caudate, fronto-occipital fasciculus, and cerebral peduncle intersections were associated with greater medication reduction; caudate intersection with greater memory decline, and other regions with apathy and impulsivity.

Introduction

Deep brain stimulation (DBS) is an effective therapy for managing Parkinson's disease (PD) symptoms, yet clinical outcomes remain variable and incompletely understood. Although lead placement within anatomical sweet spots is associated with better outcomes[1], the full lead trajectory through non-target brain structures may also contribute to outcome variability. Caudate penetration during subthalamic nucleus (STN) DBS has been associated with short-term global cognitive decline and working memory impairment,[2,3]but longer-term studies report inconsistent findings[4–7]. These inconsistencies may reflect differences in study design and warrant further investigation. Determining whether lead trajectory influences clinical response could inform surgical planning and prognosis.

Methods

We retrospectively studied 100 individuals with Parkinson's disease (mean age 64.6 ± 9.0 years, 31% female) with DBS targeting the STN (n=41) or GPi (n=59). The primary outcome was percent change in levodopa equivalent daily dose (LEDD) from baseline to 12 months (n=92); secondary 12-month outcomes were changes in MoCA, composite z-scores reflecting six cognitive domains [processing speed, attention, memory, visual-spatial function, executive function; n=11-36], and four mood measures [BDI-II (depression), BAI & PAS (anxiety), SAS (apathy), QUIP-A (impulse control); n=34-39]. A 328-region composite atlas in MNI space combined the Zhang whole-brain atlas[8], SMILE thalamic/deep grey nuclei atlas[9], and DKT gray matter cortical atlas[10] (**Fig. 1**). Leads were segmented in ITK-SNAP on postoperative CT (83%) or MRI (17%). ANTs SyN mapped the atlas into native space (**Fig. 2**). To assess localization robustness, lead masks were shifted (±1–2 mm), dilated (0.5–1.5 mm), and eroded (0.25–0.5 mm) across 25 perturbations; regions present in ≥80% perturbations were retained (**Fig. 3A**). Bootstrapped LASSO (1,000 resamples) selected regions with selection frequency >0.7 for LEDD, or the top five regions for secondary outcomes. Selected regions and covariates (age, sex, target, number of leads, baseline MDS-UPDRS III dopaminergic responsiveness) entered ridge regression with bootstrapped 95% CIs. OLS models with and without selected regions assessed added explanatory value. LEDD analyses were repeated for STN (n=37) and GPi (n=55).

Results

Selection-stable LASSO identified five stable features associated with ΔLEDD (**Fig. 3A-C**). Right cerebral peduncle (n=30; β=3.4, 95% CI [0.2, 7.3]), left superior fronto-occipital fasciculus (n=26; β=4.1, 95% CI [0.6, 9.6]) and right caudate (n=9; β=4.7, 95% CI [1.2, 9.9]) were associated with greater medication reductions after DBS. Left cerebral peduncle showed a similar trend (n=32; β=3.5, 95% CI [−2.0, 9.1]), whereas left medial forebrain bundle was associated with smaller decreases (n=11; β=−5.2, 95% CI [−15.4, 0.5]) (**Fig 4A**). Consistent with prior knowledge, STN compared to GPi DBS was associated with greater medication reductions (β=5.2, 95% CI [2.3, 10.2]; **Fig. 4B**).

Clinical covariates explained 24.8% of ΔLEDD variance, primarily driven by DBS target (β=36.9, 95% CI [20.3, 53.5], p<0.001; **Fig. 4C**). Adding stable brain features increased R^2 to 34.1%, although the improvement was not significant ($F(5,81)=2.29$, p=0.054; **Fig. 4C**). Left medial forebrain bundle was the only significant feature in this model (β=−29.3, 95% CI [−52.0, −6.5], p=0.012; **Fig. 4D**).

Target-specific analyses identified left caudal middle frontal cortex (n=30; β=10.4, 95% CI [1.3, 19.4]) and right caudate (n=8; β=10.0, 95% CI [1.8, 19.2]) associations with greater medication reductions in STN patients (n=37; **Fig. 4B**). In GPi patients (n=55), left medial forebrain bundle (n=7; β=−7.4, 95% CI [−26.8, 0.0]) and left external capsule (n=29; β=5.1, 95% CI [0.0, 15.1]) showed borderline associations (**Fig. 4B**).

Cognitive and behavioral outcomes were outcome-specific (Fig. 5). For memory (N=36, Fig. 5B), right caudate passage was associated with worsened function (n=4; β=−0.10, 95% CI [−0.25, −0.0002]). For apathy (SAS; N=35, Fig. 5C), passage of right superior longitudinal fasciculus was associated with decreased symptoms (n=8; β=−0.74, 95% CI [−1.76, −0.01]). For impulsive-compulsive behaviors (QUIP-A; N=34, Fig. 5A), five selected regions were significant: right STN (n=7; β=0.15, 95% CI [0.05, 0.24]), right superior corona radiata (n=20; β=0.14, 95% CI [0.04, 0.31]) and left middle frontal superficial white matter (n=29; β=0.10, 95% CI [0.01, 0.22]) was associated with increased impulsivity, whereas left ventrooralis posterior thalamic nucleus (n=9; β=−0.15, 95% CI [−0.32, −0.05]) and left caudate (n=8; β=−0.13, 95% CI [−0.25, −0.03]) intersection were associated with reduced impulsivity.

Discussion/Conclusion

Trajectory-associated regions explained additional outcome variability beyond DBS target, suggesting that trajectory anatomy may capture outcome-relevant differences in electrode placement. Right caudate intersection was associated with greater medication reduction but worse memory, suggesting a potential motor-cognitive trade-off; distinct regions associated with increased versus decreased impulsivity may reflect competing behavioral effects along trajectories. Given limited cognitive data and potential for imaging processing errors, findings remain exploratory. Ongoing work will refine image processing pipeline and substantially increase the cohort to assess reproducibility.

Acknowledgements

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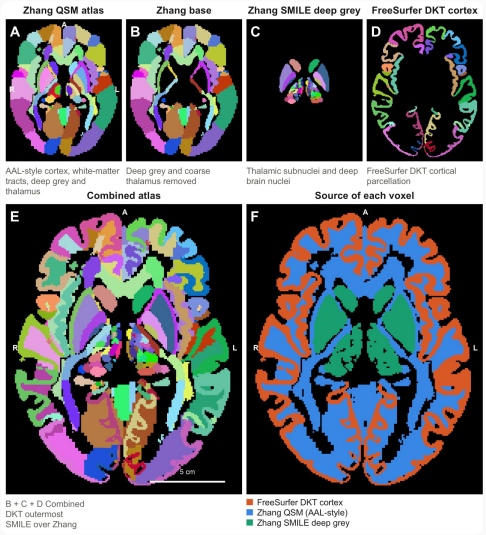
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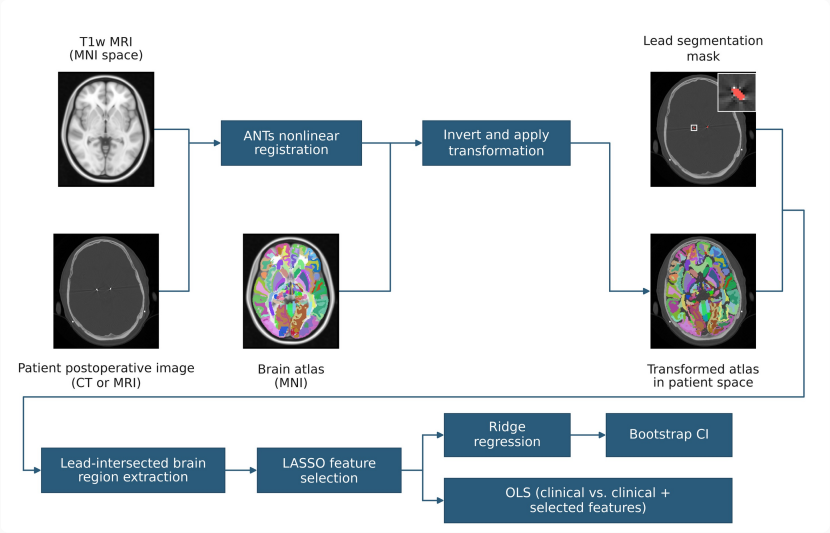
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Figures and Tables



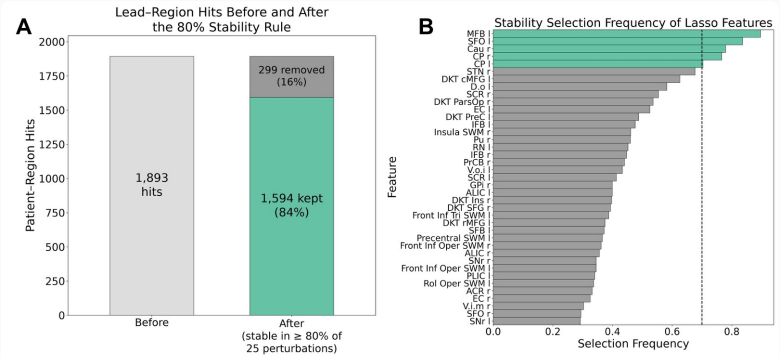
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Figure 1: Composite brain atlas. ***(A)*** *Zhang QSM atlas with AAL-style regions, white-matter tracts, deep grey matter, and thalamus.* ***(B)*** *Zhang QSM atlas with deep grey matter and coarse thalamic labels removed.* ***(C)*** *Zhang SMILE thalamic subnuclei and deep brain nuclei.* ***(D)*** *FreeSurfer DKT cortical parcellation.* ***(E)*** *Combined 328-region atlas; SMILE labels take precedence over the Zhang base, and DKT forms the outermost cortical layer.* ***(F)*** *Atlas source of each voxel.*



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Figure 2: Overview of analytical pipeline. *MNI T1w template registered to each patient's postoperative CT or MRI with ANTs nonlinear registration. Inverse transform brought the brain atlas into patient space. Regions intersected by each lead segmentation mask (inset) were extracted as binary features. Bootstrapped LASSO selected regions, which then entered ridge regression with bootstrapped confidence intervals and OLS models comparing clinical covariates alone with covariates plus the selected regions.*

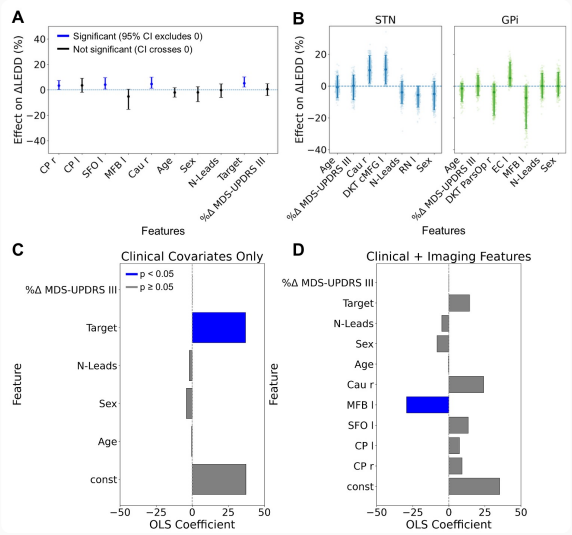


C

Step	Hits	Distinct regions
Before the stability filter (100 patients)	1,893	81
After the 80% rule (100 patients)	1,594	80
LEDD Outcomes Data (92 patients)	1,441	79 (LASSO input)
Bootstrapped LASSO Output		5

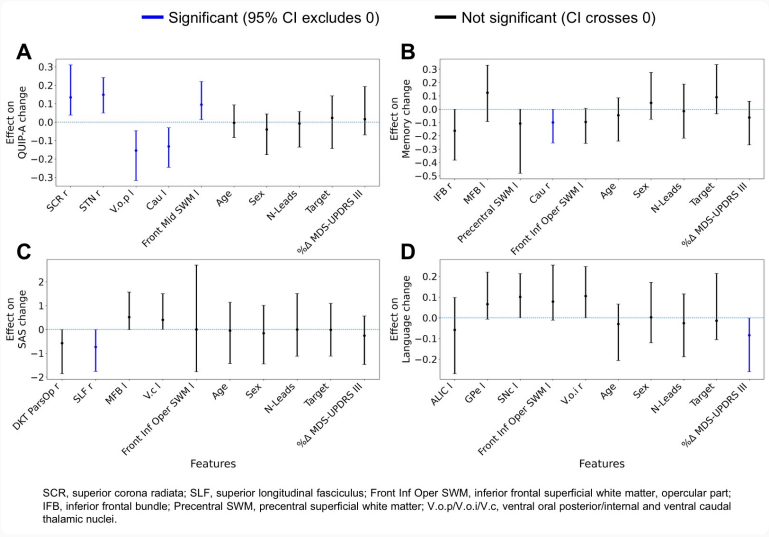
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Figure 3: Lead-region perturbation stability and stability-based LASSO feature selection and regression modeling. ***(A)*** *Lead-region overlapping hits in 100 patients before and after perturbation stability rule. A hit was retained only if present in ≥80% of 25 perturbed lead masks (shifts, dilations, erosions).* ***(B)*** *Bootstrapped LASSO selection frequency (1,000 resamples) for LEDD change. Regions above 0.70 threshold were selected (green).* ***(C)*** *Table explaining number of hits and distinct regions at each step.*



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Figure 4: Associations with LEDD change. ***(A)*** *Ridge regression coefficients (bootstrapped 95% CIs) for stability-selected regions and LEDD covariates (N=92).* ***(B)*** *Ridge regression stratified by target: STN (n=37) and GPi (n=55).* ***(C)*** *Incremental ordinary least squares (OLS) model coefficients with clinical covariates only (R^2=0.25).* ***(D)*** *OLS model with a combined clinical and imaging feature model for DLEDD (R^2=0.34).*



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Figure 5: Neuropsychiatric outcomes for the top five LASSO-selected regions and covariates: (A) *QUIP-A (n=34),* ***(B)*** *Memory (n=36)* ***(C)*** *SAS apathy (n=35),* ***(D)*** *language (n=35). Positive QUIP-A and SAS score changes (raw score based) correspond to worse outcomes, while positive memory and language score changes (z-score based) are better.*